# THE GAMMA FUNCTION

Some notes about the Gamma Function $\Gamma$

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# 1 The factorial

The factorial of a number $n$ such as $n \geq 0$ and $n \in \mathbb{N}$ is defined by the following equation:

$$n! = n(n-1)! \tag{1.1}$$

If we apply the case where $n = 1$ in the equation, we must find:

$$0! = 1 \tag{1.2}$$

Using these two properties, we can find an integral capable of expanding the factorial to all real numbers:

$$I_n = \int_0^\infty t^n e^{-t} dt = n! \tag{1.3}$$

Now we will see if the properties (1.1) and (1.2) apply:

1. Demo of $0! = 1$:

   **DEM:** $I_0 = \int_0^\infty e^{-t} dt = -e^{-t}\Big|_0^\infty = -\lim_{t\to\infty}(e^{-t}) + \lim_{t\to 0}(e^{-t}) = -0 + 1 = 1$ ♣

   This indicates that $I_0 = 0! = 1$

2. Demo of the formula $n! = n(n-1)!$

   Applying integration by parts and using the respective relations: $\begin{cases} u = t^n \to du = nt^{n-1} \\ dv = e^{-t} \to v = -e^{-t} \end{cases}$

   **DEM:** $I_n = \int_0^\infty t^n e^{-t} dt = -t^n e^{-t}\Big|_0^\infty - \int_0^\infty -nt^{n-1}e^{-t} dt = -t^{n-1}e^{-t}\Big|_0^\infty + n\int_0^\infty t^{n-1}e^{-t} dt = nI_{n-1}$ ♣

   where $\lim_{t\to\infty}(-t^{n-1}e^{-t}) = 0$, and $\lim_{t\to 0}(-t^{n-1}e^{-t}) = 0$

# 2 Gamma Function

In the integral $I_n$, $n$ can be any real number and only converge if $n \geq 0$. This information can be helpful to understand the definition of the *Gamma Function*, which is:

$$\Gamma(x) = \int_0^\infty t^{x-1} e^{-t} dt \tag{2.1}$$

$\Gamma(x)$ is defined as $\forall x > 0$. When $x = n \in \mathbb{N}$, $\Gamma(n) = (n-1)!$. Therefore, $n! = \Gamma(n+1)$. Now we can extend the factorial for any real number $x \geq 0$:

$$\Gamma(x+1) = x! \tag{2.2}$$

And, for $x > 0$:

$$\Gamma(x) = (x-1)! \tag{2.3}$$

Using the demo previously seen, we obtain the following:

$$\Gamma(x+1) = x\Gamma(x) \tag{2.4}$$

or

$$\Gamma(x) = (x-1)\Gamma(x-1) \tag{2.5}$$

The integral equation (2.1) can not be expressed as a finite sum of elemental functions when $x \notin \mathbb{N}$. Because of that, there is a function similar to (2.1), which values only can be obtained by mathematical software. The property (2.5) is enough to have the values of the *Gamma Function* for $1 < x < 2$.

It is quite easy to see that $\Gamma(x)$ as a function verifies:

$$\lim_{x\to 0^+} \Gamma(x) = \infty, \qquad \lim_{x\to\infty} \Gamma(x) = \infty \tag{2.6}$$

<u>Observation:</u> By doing $x \to 0$ in (2.4) we obtain $x\Gamma(x) \sim \Gamma(1)$, then $\Gamma(x) \sim \dfrac{1}{x}$ with $x \to 0$.

$\Gamma(x)$ has a minimum in $x = 1.4616...$ and its value is $0.8856....$

The *Gamma Function* can be generalized to a complex variable function $\Gamma(s)$, $s \in \mathbb{C}$ with $s \neq 0, -1, -2, \cdots$. At non-positive integers the module of the *Gamma Function* tends to infinity.

## 2.1 The reflection formula

For $0 < x < 1$ :

$$\Gamma(x)\Gamma(1-x) = \frac{\pi}{\sin \pi x} \tag{2.7}$$

If $x = \dfrac{1}{2}$, then $\Gamma\left(\dfrac{1}{2}\right)^2 = \pi$, where:

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi} \tag{2.8}$$

Using the properties seen before (2.5) and (2.8), we can obtain a formula for $x = \frac{n}{2}$, $n \in \mathbb{N}$:

- If n is odd,

$$\Gamma\left(\frac{n}{2}\right) = \frac{(n-2)!!}{2^{\frac{n-1}{2}}}\sqrt{\pi} \tag{2.9}$$

  where the double factorial of an odd number is defined as $n!! = 1 \cdot 3 \cdot 5 \cdots$

- If n is even, the formula is simply

$$\Gamma\left(\frac{n}{2}\right) = \frac{n}{2} - 1 \tag{2.10}$$

## 2.2 Application to integral calculous

There are a lot of integrals that can be expressed in terms of the *Gamma Function*. Let's see some examples.

$$\int_0^\infty x^a e^{-\lambda x} dx = \frac{\Gamma(a+1)}{\lambda^{a+1}}. \quad (a > -1),\ (\lambda > 0) \tag{2.11}$$

**DEM:** Using the change $t = \lambda x$, $dt = \lambda dx$:

$$\int_0^\infty x^a e^{-\lambda x} dx = \int_0^\infty \left(\frac{t}{\lambda}\right)^a e^{-t} \frac{dt}{\lambda} = \int_0^\infty \frac{t^a}{\lambda^{a+1}} e^{-t} dt = \frac{\Gamma(a+1)}{\lambda^{a+1}} \quad \clubsuit$$

$$\int_0^\infty x^a e^{-x^2} dx = \frac{1}{2}\Gamma\Big(\frac{a+1}{2}\Big). \quad (a > -1) \tag{2.12}$$

**DEM:** Using the change $t = x^2$, $x = \sqrt{t}$ $dx = \dfrac{dt}{2\sqrt{t}}$:

$$\int_0^\infty x^a e^{-x^2} dx = \int_0^\infty t^{\frac{a}{2}} e^{-t} \frac{dt}{2\sqrt{t}} = \frac{1}{2}\int_0^\infty t^{\frac{a-1}{2}} e^{-t} dt = \frac{1}{2}\Gamma\Big(\frac{a+1}{2}\Big) \quad \clubsuit$$

Using this expression we can obtain the very important value of the *Gaussian Integral*:

$$\int_{-\infty}^\infty e^{-x^2} dx = \sqrt{\pi} \tag{2.13}$$

**DEM:** $\displaystyle\int_{-\infty}^\infty e^{-x^2} dx = 2\int_0^\infty e^{-x^2} dx = 2\frac{1}{2}\Gamma\Big(\frac{1}{2}\Big) = \sqrt{\pi} \quad \clubsuit$

## 2.3 Stirling's Formula

x! tends to infinite when $x \to \infty$. The next result shows up an equivalent infinite when $x \to \infty$:

$$x! \sim \sqrt{2\pi x} \cdot x^x e^{-x} \tag{2.14}$$

The equivalence becomes useful to study the asymptotic behavior of expressions with factorials. Besides, it is a good approximation to calculate the factorial of huge numbers.

**DEM:** Let's see the calculus without detail until:

$$x! = \int_0^\infty t^x e^{-t} dt = \int_0^\infty e^{x \ln t - t} dt.$$

The exponent has the derivatives: $\dfrac{d}{dt}(x \ln t - t) = \dfrac{x}{t} - 1$, $\dfrac{d^2}{dt^2}(x \ln t - t) = -\dfrac{x}{t^2}$. Therefore, the maximum exponent is in $t = x$. The integral is refereed to this maximum so it is convenient to do the variable change $t = x + y$, $dt = dy$:

$$x! = \int_{-x}^\infty e^{x \ln(x+y) - x - y}\, dy = \int_{-x}^\infty e^{x \ln x + x \ln(1+\frac{y}{x}) - x - y}\, dy = e^{x \ln x - x} \int_{-x}^\infty e^{x \ln(1+\frac{y}{x}) - y}\, dy.$$

Now we have to expand the *Taylor series* of the exponent around $x = 0$:

$$x! = x^x e^{-x} \int_{-x}^\infty e^{-\frac{y^2}{2x} + \frac{y^3}{3x^2} - \frac{y^4}{4x^3} + \cdots}\, dy$$

Finally, with $y = \sqrt{2x} \cdot z$, $dy = \sqrt{2x} \cdot dz$ and $x \to \infty$:

$$x! = \sqrt{2x}\, x^x e^{-x} \int_{-x}^\infty e^{-z^2 + \frac{2\sqrt{2}\, y^3}{3\sqrt{x}} - \frac{z^4}{x} + \cdots}\, dz \sim \sqrt{2x}\, x^x e^{-x} \int_{-\infty}^\infty e^{-z^2} dz = \sqrt{2x}\, x^x e^{-x} \sqrt{\pi} \quad \clubsuit$$

# 3 Relation between Gamma function and Riemann function

The Riemann $\zeta$ function is one of the most studied functions in mathematics not only by its relation with one of the *millennium problems,* but for its applications in the prime number's theory.

The Riemann function is as follows:

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s} \tag{3.1}$$

We will see how this function, which is a sum of infinite elements, becomes an integral by using the *Gamma Function.* To do this, let us start with the following integral.

$$\int_0^{\infty} x^{s-1} e^{-nx}\, dx = \frac{\Gamma(s)}{n^s} \tag{3.2}$$

**DEM:** With the variable change $t = nx$, $dt = n\, dx$

$$\int_0^{\infty} x^{s-1} e^{-nx}\, dx = n \int_0^{\infty} \left(\frac{t}{n}\right)^{s-1} e^{-t}\, dt = \frac{1}{n^s} \int_0^{\infty} t^{s-1} e^{-t}\, dt = \frac{\Gamma(s)}{n^s} \quad \clubsuit$$

If we do the product between $\Gamma(s)$ (expressed by the equation (3.2)) and $\zeta(s)$:

$$\Gamma(s)\zeta(s) = n^s \Big(\sum_{n=1}^{\infty} \frac{1}{n^s}\Big) \int_0^{\infty} x^{s-1} e^{-nx} dx = \sum_{n=1}^{\infty} \int_0^{\infty} x^{s-1} e^{-nx} dx = \int_0^{\infty} x^{s-1} \Big(\sum_{n=1}^{\infty} e^{-nx}\Big) dx = \int_0^{\infty} \frac{x^{s-1}}{e^x - 1} dx\,,$$

where $\displaystyle\sum_{n=1}^{\infty} e^{-nx} = \sum_{n=1}^{\infty} (e^{-x})^n = \frac{e^{-x}}{1 - e^{-x}} = \frac{1}{e^x - 1}$.

Looking at the expression recently found, we can express the *Riemann function* in terms of the *Gamma function.*

$$\zeta(s) = \frac{1}{\Gamma(s)} \int_0^{\infty} \frac{x^{s-1}}{e^x - 1}\, dx \tag{3.3}$$

This is one method to analytically extend the *Riemann function* to the entire complex plane, giving birth to solutions such as: $\zeta(-1) = \displaystyle\sum_{n=1}^{\infty} n = -\frac{1}{12}$.